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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The rapid development of digital technologies and artificial intelligence has significantly transformed many aspects of modern life, including the fashion industry. Digital wardrobe systems have emerged as innovative solutions that enable individuals to organize, manage, and visualize their clothing collections digitally. These systems allow users to upload images of their clothing items and manage them in an organized manner, helping them make better outfit decisions and maximize the use of their wardrobe.

However, most existing digital wardrobe applications require users to manually enter clothing details such as type, color, and style. This manual labeling process can be time-consuming and inconvenient, especially for users who have large clothing collections. In addition, manually entering clothing attributes may lead to inconsistent data and errors.

Recent advancements in artificial intelligence and computer vision have made it possible to automate image recognition tasks using deep learning techniques. Convolutional Neural Networks (CNNs) have demonstrated exceptional performance in image classification problems, including object detection and visual attribute recognition.

This research proposes an AI-powered clothing classification module that automatically identifies and categorizes clothing items from images uploaded by users. The proposed system integrates an automated background removal technique to isolate clothing items from complex backgrounds before performing classification. The cleaned image is then processed using a deep learning model to extract visual features and classify the clothing item based on multiple attributes such as clothing type, color, and pattern.

The results of the classification process are stored in a digital wardrobe database, enabling users to easily organize and search their clothing items. The system also provides structured data that can be used by other modules of the digital wardrobe system, such as outfit recommendation engines.

The performance of the proposed model will be evaluated using common classification metrics including accuracy, precision, recall, and F1-score. The proposed solution aims to improve the usability and efficiency of digital wardrobe systems by reducing manual data entry and enhancing clothing organization through intelligent automation.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
CNN	Convolutional Neural Network
CV	Computer Vision
U^2-Net	U-Square Net Background Removal Model
RGB	Red Green Blue
API	Application Programming Interface
SQL	Structured Query Language
GPU	Graphics Processing Unit
ML	Machine Learning
NLP	Natural Language Processing

1 INTRODUCTION

1.1 Background

The fashion industry has undergone a significant transformation with the integration of digital technologies and artificial intelligence. In modern society, individuals often own a large number of clothing items, which makes managing wardrobes increasingly challenging. People frequently struggle to remember all the clothing items they own, leading to inefficient outfit selection and underutilization of garments.

Digital wardrobe systems have emerged as a practical solution to address this issue. These systems allow users to create a virtual representation of their wardrobe by uploading images of their clothing items and organizing them digitally. Users can browse their wardrobe collections through a mobile application or web platform, making it easier to plan outfits and manage clothing inventory.

Despite the benefits of digital wardrobe systems, many existing platforms rely heavily on manual data entry. Users are required to manually categorize clothing items by specifying attributes such as clothing type, color, fabric, and style. This process can be time-consuming and may discourage users from fully utilizing the system.

Artificial intelligence, particularly computer vision and deep learning, provides an opportunity to automate the process of clothing recognition. Computer vision techniques enable machines to analyze and interpret visual information from images. Deep learning models, such as Convolutional Neural Networks (CNNs), have proven highly effective in identifying objects and extracting visual features from images.

In the context of fashion technology, CNN models can be used to classify clothing images and automatically determine attributes such as clothing category, color, and pattern. This automation significantly reduces the need for manual input and improves the overall user experience of digital wardrobe systems.

However, one of the major challenges in clothing classification is the presence of complex backgrounds in user-uploaded images. Clothing images taken in real-life environments often include surrounding objects, shadows, and lighting variations that may interfere with the classification process.

To overcome this challenge, background removal techniques can be used to isolate the clothing item from the background before classification. By removing unnecessary visual elements, the classification model can focus on the clothing item itself, resulting in improved accuracy.

This research aims to develop an intelligent clothing classification module that integrates background removal and deep learning techniques to automatically categorize clothing images within a digital wardrobe system.

In addition to intelligent clothing classification, the proposed system incorporates a trend-aware module that leverages Google Trends data to track emerging fashion patterns. By analyzing search queries and popular keywords related to clothing types, colors, fabrics, and styles, the system can identify trending fashion elements in real time. This trend information is then compared with the user's existing wardrobe to detect items that align with current fashion trends. When matches are found, the system can provide personalized suggestions, enabling users to create stylish outfits without acquiring new garments. By bridging personal wardrobe management with dynamic fashion trends, this feature enhances user engagement, supports informed styling decisions, and demonstrates the practical potential of combining AI-driven classification with trend analysis in modern digital wardrobe systems.

1.1.1 Statistical Evidence

The increasing popularity of digital wardrobe systems and fashion technology applications is supported by several industry reports and academic studies.

According to recent global fashion industry reports, the fashion technology market has experienced significant growth due to the integration of artificial intelligence and computer vision technologies. Studies indicate that more than 60% of consumers experience difficulty in managing their clothing collections, often forgetting items they already own or struggling to create outfit combinations.

In addition, surveys conducted among digital wardrobe application users reveal that over 70% of users prefer automated clothing recognition features instead of manually labeling clothing attributes such as color, type, and pattern.

Research also shows that the global fashion technology market is expected to exceed USD 10 billion by 2030, driven by the adoption of AI-based solutions in wardrobe management, fashion recommendation systems, and personalized styling platforms.

Furthermore, datasets such as DeepFashion, which contains over 800,000 clothing images with attribute annotations, demonstrate the growing importance of machine learning approaches in clothing classification research.

These statistics highlight the increasing demand for intelligent wardrobe management systems that can automate clothing recognition and assist users in organizing their wardrobes efficiently.

1.2 Literature Survey

The application of computer vision and machine learning in the fashion industry has attracted significant research attention in recent years. Researchers have explored various techniques for clothing recognition, attribute detection, and fashion recommendation systems.

One of the most widely used datasets in fashion image research is the DeepFashion dataset, which contains a large number of clothing images with annotated attributes. The dataset includes various clothing categories such as shirts, dresses, pants, jackets, and skirts. It has been widely used to train deep learning models for fashion image recognition.

Convolutional Neural Networks (CNNs) have become the dominant approach for image classification tasks. CNN architecture such as VGGNet, ResNet, and MobileNet have demonstrated high performance in recognizing visual patterns and features. These models use multiple convolutional layers to extract hierarchical features from images, enabling them to identify complex patterns such as textures, shapes, and colors.

Several researchers have applied CNN-based models to clothing classification tasks. These models can automatically detect clothing attributes such as sleeve length, collar type, and clothing style. The use of deep learning has significantly improved the accuracy of fashion recognition systems compared to traditional machine learning methods.

Another important area of research is image segmentation and background removal. Segmentation techniques divide an image into different regions, allowing the system to isolate objects of interest. Modern deep learning-based segmentation models, such as U^2-Net, are capable of generating accurate foreground masks that separate objects from the background.

Background removal is particularly important in clothing classification because background elements may introduce noise into the feature extraction process. By removing the background, the model can focus on the clothing item itself and extract more relevant features.

Recent research has also explored multi-attribute classification in fashion images. Instead of predicting a single category, multi-attribute models can simultaneously predict multiple clothing attributes such as category, color, and pattern. This approach provides more detailed information about clothing items and enhances the functionality of digital wardrobe systems.

1.3 Research Gap

Although numerous studies have investigated fashion image recognition, clothing classification, and digital wardrobe systems, several important limitations remain unresolved.

First, many existing digital wardrobe applications require users to manually enter clothing attributes such as type, color, and style. This manual labeling process is inefficient and can become tedious when users manage large clothing collections. As a result, many users abandon digital wardrobe applications due to the effort required for manual data entry.

Second, most existing clothing classification models are trained using curated datasets where images contain minimal background noise. In real-world scenarios, however, users typically capture clothing images using mobile devices in uncontrolled environments where complex backgrounds, shadows, and lighting variations are present. These conditions can significantly reduce the accuracy of image classification models.

Third, many existing research approaches focus on predicting a **single clothing attribute**, such as clothing category. However, practical wardrobe management systems require **multi-attribute classification**, including attributes such as clothing type, color, and pattern in order to organize wardrobe items effectively.

Furthermore, although recent studies have explored fashion recommendation systems, **limited research has integrated clothing classification with real-time fashion trend analysis**. The ability to combine personal wardrobe data with current fashion trends can significantly enhance user engagement and improve outfit recommendations.

Therefore, there is a need for an intelligent system that integrates **background removal, multi-attribute clothing classification, and fashion trend analysis** to improve the usability and functionality of digital wardrobe systems.

1.4 Research Problem

The main problem addressed in this research is the lack of automated clothing classification in digital wardrobe systems.

Users are currently required to manually categorize clothing items, which can be time-consuming and inconvenient.

Additionally, the presence of background noise in clothing images can reduce the accuracy of classification models.

This research aims to develop an AI-based clothing classification module that automatically identifies clothing attributes from images while minimizing the impact of background noise through background removal techniques.

2. OBJECTIVES

Main Objective

To develop an AI-based clothing classification module that automatically categorizes clothing images for a digital wardrobe system.

Specific Objectives

- Implement an automated background removal system for clothing images to enhance image preprocessing for downstream tasks.
- Develop a CNN-based model for clothing attribute classification, enabling accurate recognition of clothing features.
- Classify clothing items based on multiple attributes such as type, color, and pattern to support detailed wardrobe organization.
- Integrate classification results with a digital wardrobe database to enable efficient retrieval, management, and outfit recommendation.

3. METHODOLOGY

The Smart Wardrobe Organization System consists of several interconnected modules that process clothing images and organize them within a digital wardrobe environment. The architecture is designed to automate the classification of clothing items while maintaining high accuracy and efficiency.

The system begins with the image acquisition stage, where the user uploads a clothing image through the application interface. This image is then processed by the background removal module, which isolates the clothing item from surrounding objects in the image.

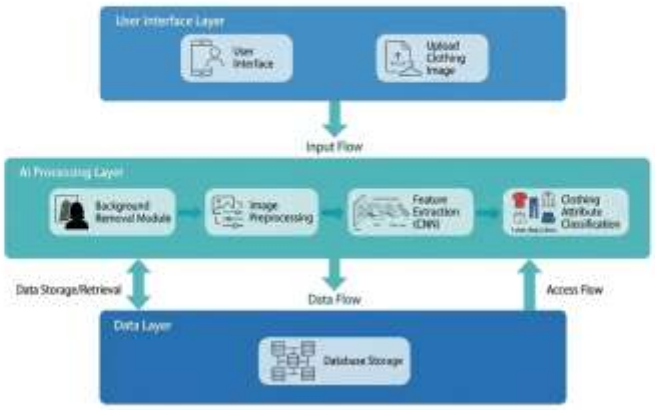
After background removal, the image is passed to the image preprocessing module, where operations such as resizing, normalization, and noise reduction are performed to improve the input quality.

The processed image is then sent to the Convolutional Neural Network (CNN) feature extraction module, which identifies visual patterns such as shapes, colors, and textures.

The extracted features are used by the multi-attribute classification module to determine different attributes of the clothing item including type, color, and pattern.

Finally, the classified information is stored in the wardrobe database, which organizes the clothing items for easy retrieval and management. The database also interacts with the Google Trends analysis module, which monitors current fashion trends and provides notifications to users if their wardrobe items match trending styles.

Diagram 1 – System Architecture



Overall System Architecture

3.1 Image Acquisition

Users upload clothing images through the digital wardrobe application. These images may be captured using mobile phone cameras or selected from the user's image gallery.

The uploaded images serve as the input to the clothing classification module.

3.2 Background Removal

Background removal is performed to isolate the clothing item from its surrounding environment.

Deep learning-based segmentation models are used to identify the foreground clothing object and remove background elements.

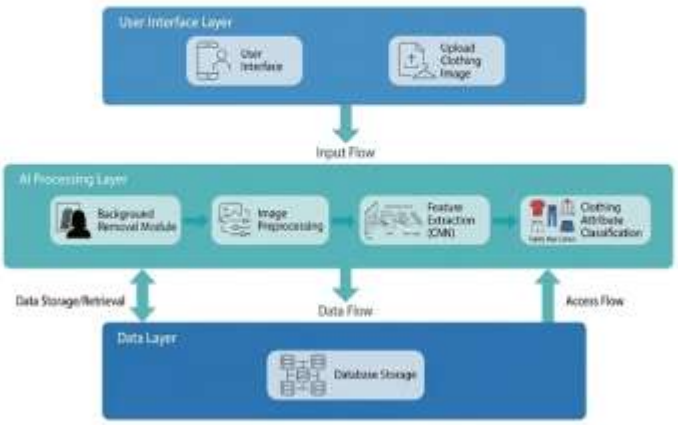
3.3 Image Preprocessing

Before feeding the images into the classification model, several preprocessing steps are applied.

These steps include:

- Image resizing
  Normalization
  Noise reduction
  Color adjustment

Preprocessing ensures that all images have consistent dimensions and quality.



3.4 Feature Extraction Using CNN

Convolutional Neural Networks automatically extract visual features from images.

CNN models consist of several layers including convolution layers, pooling layers, and fully connected layers.

The Convolutional Neural Network (CNN) architecture is used to extract meaningful visual features from clothing images. CNN models are particularly effective for image classification tasks because they can automatically learn hierarchical feature representations.

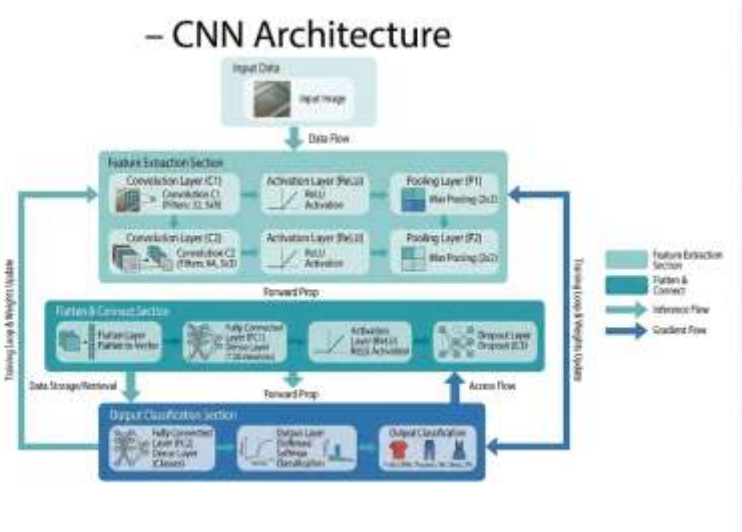
The architecture typically consists of several layers including convolution layers, pooling layers, and fully connected layers.

The **convolution layers** apply filters to the input image in order to detect important visual features such as edges, shapes, and textures.

The **pooling layers** reduce the spatial dimensions of the feature maps while retaining the most important information. This helps reduce computational complexity and prevents overfitting.

The extracted features are then passed to **fully connected layers**, which perform the final classification of clothing attributes such as clothing type, color, and pattern.

Diagram - CNN Architecture



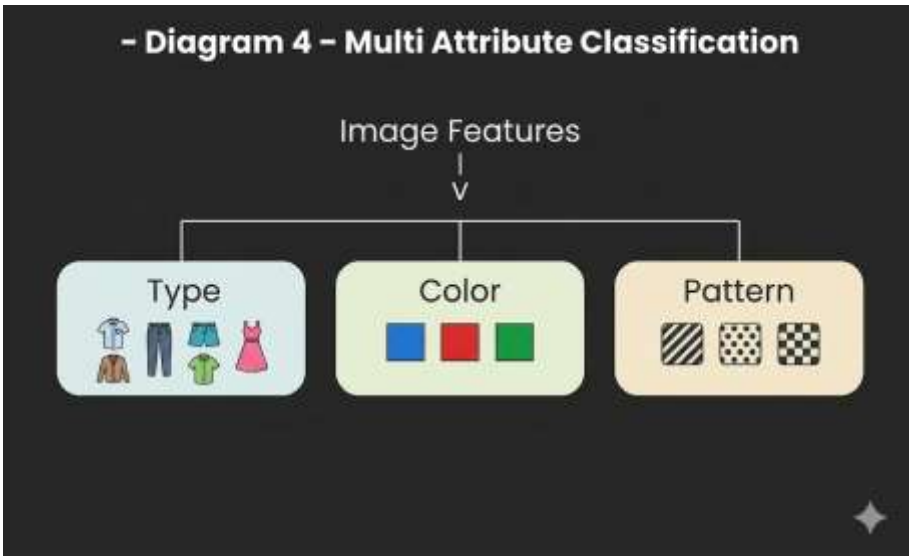
3.5 Clothing Classification

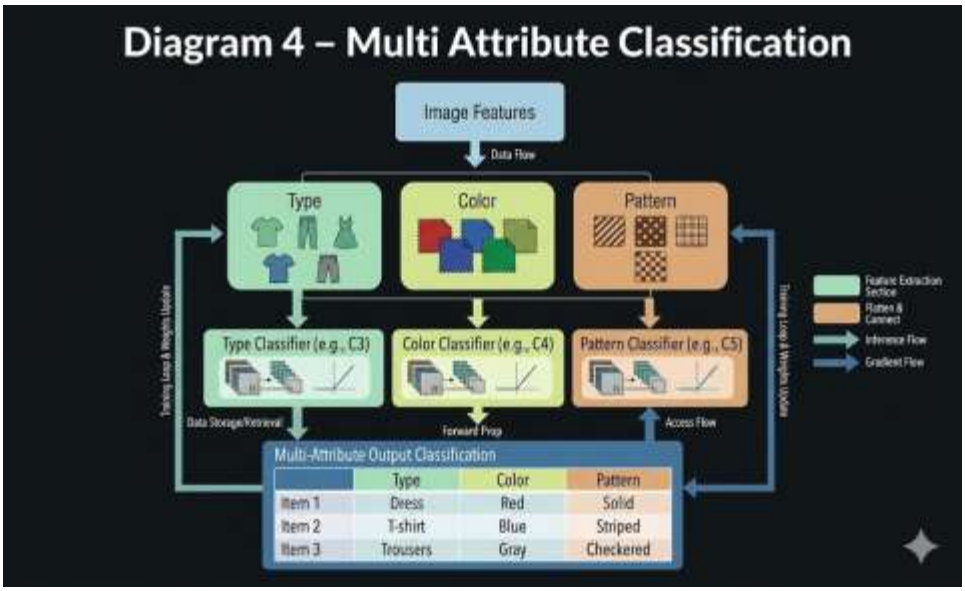
The classification model predicts multiple attributes for each clothing item.

Examples:

Clothing Type
Color
Pattern

Diagram – Multi Attribute Classification





3.6 Google Trends Fashion Analysis

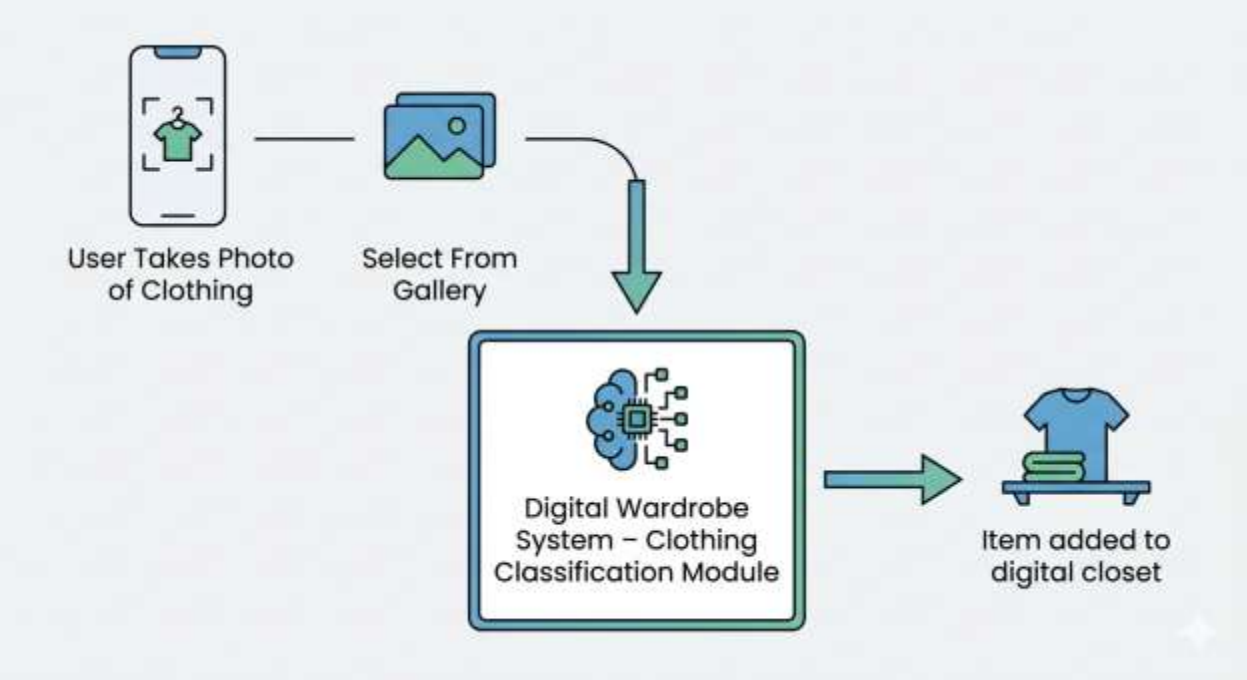
Google Trends is an online tool that analyzes the popularity of search queries across different regions and time periods. By examining fashion-related search trends, it is possible to identify clothing styles that are gaining popularity among users.

In the proposed system, Google Trends data will be used to monitor popular clothing items and fashion styles. The system periodically retrieves search trend information related to clothing categories such as jackets, hoodies, dresses, and streetwear.

The collected trend data is analyzed to determine which clothing styles are currently trending in the fashion industry. This information is then compared with the clothing items stored in the user's wardrobe database.

If a clothing item in the user's wardrobe matches a trending fashion style, the system will generate a notification suggesting that the user can wear that item as part of a fashionable outfit.

This feature provides users with valuable fashion insights while encouraging them to make better use of their existing wardrobe items.



3.7 Wardrobe Scheduling and Notification Module

In addition to clothing classification and trend analysis, the proposed system includes a wardrobe scheduling and reminder module to enhance user convenience and planning.

This module allows users to schedule outfits based on their upcoming events, functions, or daily activities. Users can select clothing items from their digital wardrobe and assign them to specific dates and occasions such as parties, meetings, weddings, or casual outings. This helps users plan their outfits in advance and reduces the time spent on daily outfit selection.

The scheduling data is stored in the system database along with event details such as event name, date, time, and selected clothing items. The system maintains a structured calendar-like view where users can easily manage and review their planned outfits.

Furthermore, the system integrates a notification mechanism that sends reminders to users about their scheduled outfits. Notifications are generated prior to the scheduled event time, ensuring that users are reminded to wear their planned outfit. These reminders can be delivered through mobile notifications or in-app alerts.

This feature improves user experience by combining wardrobe management with personal scheduling, enabling better organization, time efficiency, and effective utilization of clothing items.

3.8 Database Storage

After classification, the results are stored in the digital wardrobe database.

Example database fields:

Clothing ID
Image
Type
Color
Pattern
Date Added

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

Functional requirements describe the main operations and tasks that the system must perform.

Upload Clothing Images
The system should allow users to upload images of their clothing items through the application interface. These images may be captured using a mobile camera or selected from the device gallery.

Remove Background Automatically
The system should automatically remove the background from uploaded images in order to isolate the clothing item. This helps improve the accuracy of the classification process.

Classify Clothing Attributes
The system should analyze the clothing image using a deep learning model and classify the item based on attributes such as clothing type, color, and pattern.

Store Results in Database
After classification, the system should store the clothing image and its detected attributes in the digital wardrobe database so that users can easily organize and access their clothing items.

Scheduling Clothing's

The system should allow users to schedule outfits based on upcoming events and send reminder notifications accordingly.

4.2 Non-Functional Requirements

Non-functional requirements describe the quality and performance characteristics of the system.

High Accuracy

The classification model should provide accurate predictions when identifying clothing attributes to ensure reliable organization of wardrobe items.

Fast Processing

The system should process uploaded images quickly so that users receive classification results without long delays.

User-Friendly Interface

The application should provide a simple and easy-to-use interface that allows users to upload images, view classified clothing items, and manage their digital wardrobe without difficulty.

4.5 Budget Estimation

The development of the Smart Wardrobe Organization System requires several software tools, cloud services, and hardware resources. These resources are necessary for developing the mobile application, implementing machine learning models, performing image processing tasks, and storing project data.

The software costs include development environments, machine learning frameworks, AR development tools, and supporting libraries required for mobile application development, deep learning model implementation, and image processing. Although some tools such as Python, Blender, and Unity provide free versions, additional plugins, development utilities, and assets may require licensing or subscription payments.

Cloud services are required for model storage, data hosting, and system deployment. Internet connectivity is also essential for downloading datasets, accessing cloud services, and collaborating with development platforms.

The estimated budget required for a six-month development period is presented in Table 4.1.

Item	**Estimated Amount for a 6-Month Period (SL Rupees)**
Licensed Software Tools	10,000.00
Cloud Services / Hosting	10,000.00
Online Courses / Workshops	10,000.00
External Storage	10,000.00
Internet / Networking	12,000.00
Other Hardware	10,000.00
Total	**62,000.00**

Diagram – Work Breakdown Structure

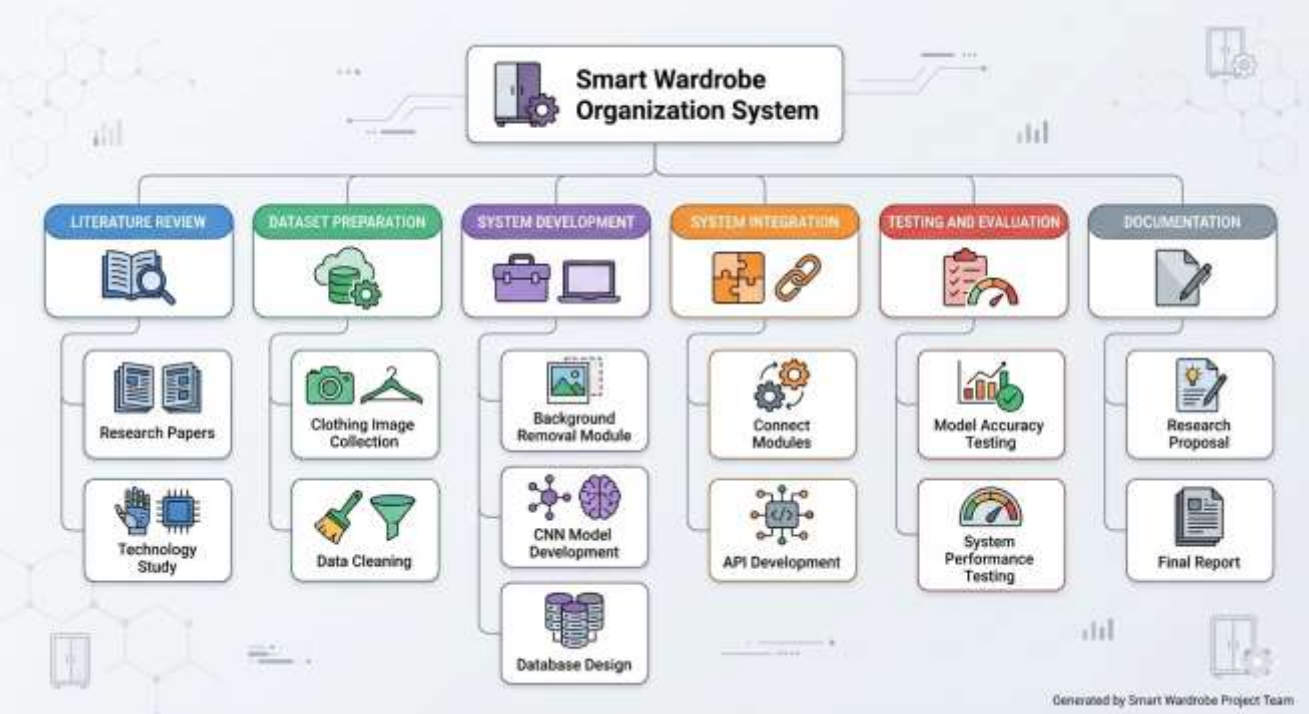
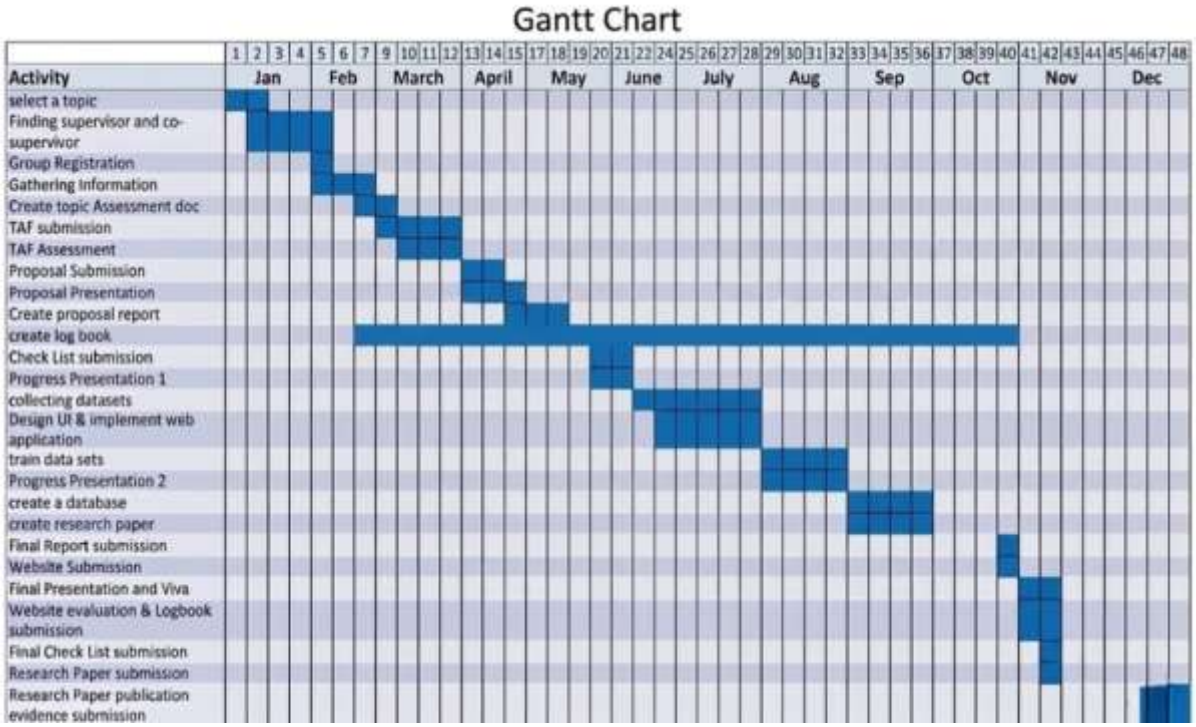


Diagram – Gantt Chart



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